



TCET THAKUR COLLEGE OF
ENGINEERING & TECHNOLOGY

Approved by AICTE & Govt. of Maharashtra | Affiliated to University of Mumbai (UoM)

- Conferred Autonomous Status by UGC for 10 years w.e.f. A.Y. 2019-20
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SANKET

AI-Powered Real-Time First-Aid Guidance for Medical Emergencies

Edge-AI Dual Engine • Visual Triage + On-Device RAG Reasoning



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OVERVIEW

Abstract

SANKET is a dual-engine mobile app built to fight the temporal compression that exhausts the **"golden hour"** in medical emergencies.

- Combines on-device visual triage (lightweight CV) with a reasoning layer powered by quantized Small Language Models (SLMs)
- Executes Retrieval-Augmented Generation (RAG) locally — fits within a strict 2GB Android memory cap
- Uses model compression (GPTQ, QLoRA) + multilingual embeddings for localized instructions
- Goal: zero-hallucination, instant first-aid guidance for untrained bystanders before responders arrive

2GB

Mobile RAM budget

Local

RAG — offline capable

0%

Hallucination target



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THE PROBLEM

Problem Statement

In mass casualty incidents or remote emergencies, delayed first aid directly increases mortality and long-term disability. Existing approaches fall short in two directions:

Traditional Manual Triage

- Static, paper-based methods
- No continuous monitoring
- Prone to human error in chaotic settings

Cloud-Dependent AI

- Fails during network outages / infrastructure collapse
- Exactly when guidance is needed most
- Transmitting live health data risks privacy & compliance violations



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LITERATURE-BACKED NEED

Research Gap

Core tension: edge-device computational limits vs. the need for safe, near - zero-hallucination medical guidance

1

Mobile edge devices have strict memory budgets, CPU thermal throttling, and fast battery drain — running large LLMs is impossible

2

3-bit / 4-bit quantization fits models on-phone, but reduced precision can degrade reasoning accuracy

3

Standard SLMs lack deep medical knowledge and are prone to hallucination

4

On-device RAG grounds answers in verified manuals, but long-context prefill is slow — historically up to 14.2s to first token



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WHY IT MATTERS

Societal Impact

Deploying SANKET builds a resilient, decentralized emergency framework — critical for Low- and Middle-Income Countries (LMICs) and mixed urban-rural settings.



Edge Processing

Data processed on-device, bypassing fragmented care pathways and lack of centralized specialists



Multilingual Reach

Supports up to multi languages/dialects for localized vernacular instructions



Power to Laypersons

Shifts emergency response from constrained institutions into hands of bystanders & community health workers



Health Equity

Drives progress toward universal health coverage in underserved regions



Literature Survey

Title	Year	Dataset	Key Inference
Pocket RAG: On-Device RAG for First Aid Guidance	2026	WHO Basic Emergency Care & Psychological First Aid	94.5% physical / 97.0% psych. first-aid accuracy; TTFT cut 14.2s → 3.7s under 2GB RAM
AI-powered AR glasses for first-aid guidance	2025	Custom Roboflow (Bleeding, Burns, CPR Chest)	YOLOv5 on Jetson Nano gives accurate real-time AR guidance without cloud dependency
QLoRA: Efficient Finetuning of Quantized LLMs	2023	OASST1, FLAN v2, Alpaca	4-bit NF4 double quantization finetunes 65B models on one GPU at full 16-bit performance
GPTQ: Accurate Post-Training Quantization	2023	WikiText2, PTB, C4	Second-order quantization compresses 175B models to 3–4 bits with negligible loss
Phi-3 Technical Report	2024	Filtered web + synthetic data	3.8B model runs natively on iPhone A16 at >12 tok/s, matching GPT-3.5 quality
Qwen3 Technical Report	2025	36T tokens, multi-languages	Strong-to-weak distillation yields edge SLMs (0.6B–32B) with unified reasoning modes
M3-Embedding: Multi-Linguality, Multi-Functionality	2024	MIRACL, MKQA, MLDR, MultiLongDoc	Versatile embedding model spans 100+ languages for cross-lingual dense/sparse retrieval



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ARCHITECTURE

Proposed Work — Models

ENGINE 1 · VISION

YOLOv5s

Lightweight, single-stage object detection — chosen over YOLOv11 for mobile fit

- Real-time visual triage & anatomical localization (e.g., CPR chest target)
- Burn severity classification
- ~1.9–2.6M parameters — stays well under 2GB RAM budget
- Millisecond inference — resists thermal throttling in field conditions
- Mature .tflite / CoreML export pipeline for stable mobile deployment

ENGINE 2 · REASONING

Qwen3 / Phi-3.5-Vision

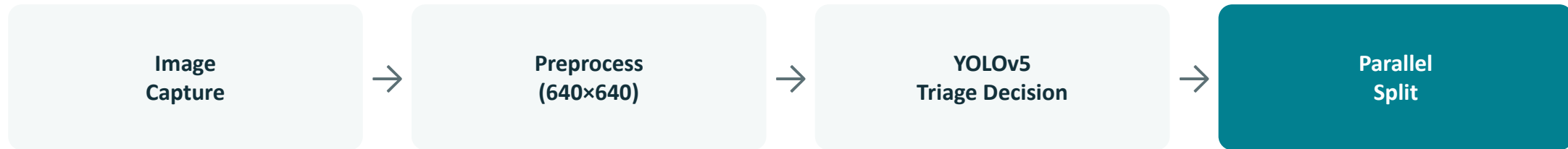
Quantized SLMs with on-device Hybrid RAG

- Qwen3 (0.6B–4B) / Phi-3.5-Vision (4.2B) — text + image tokens
- RAG grounded in local vector index of WHO emergency manuals
- 4-bit quantization: GPTQ / QLoRA NF4 + Double Quantization
- LoRA fine-tune: rank $r=64$, $\alpha=16$, dropout 0.1
- 8-bit KV Cache Quantization doubles bandwidth in RAG prefill



EXECUTION LOGIC

Flowchart & Algorithm



Thread A — Low-Latency Alert

- Trigger AR bounding boxes on-screen
- Fire asynchronous background alert to nearby verified medical and health workers
- Broadcast geofencing payload to EMS command centre

Thread B — Generative Instruction Stream

- Query passed to Hybrid On-Device RAG
- Lexical + semantic retrieval pulls correct WHO protocol
- Quantized SLM streams steps via batched decoding
- Steps pushed to Text-to-Speech (TTS) module



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User Interface

Built for extreme cognitive stress — high scannability, single-action routing, hands-free guidance.

1

Single-Action Selection

Large-icon home buttons for exact scenarios: Chest (CPR), Wound, Burn, First Aid Kit

2

High-Contrast Checklists

Discrete white cards on dark background; steps like "Apply firm pressure" shown one at a time

3

Hands-Free Voice

Every step paired with a Play button — TTS engine (pyttsx3) reads instructions aloud

4

Live AR Coaching (Future Work)

Camera overlay highlights target areas, e.g. chest-centre marker for CPR placement



SANKET

AI-powered first aid & emergency response

Welcome back

Email

you@example.com

Password

.....

Sign In

New here? [Create an account](#)

Welcome,

himeshpal658@gmail.com



Report Emergency



Injury Guidance

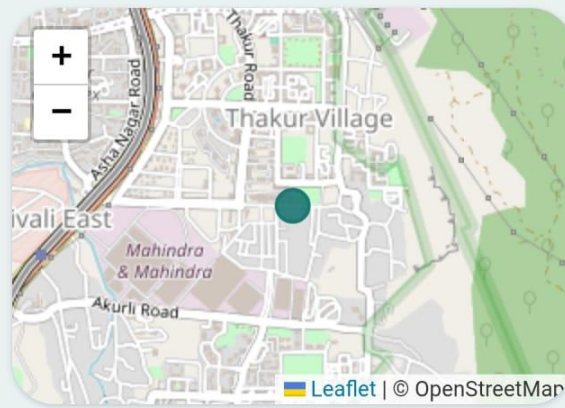
Step-by-step first aid



CPR Guide

Hands-only, 110 BPM

Nearby



0 active incident(s) nearby



Home



CPR Guide



Profile

CPR Guide

Hands-only CPR at 110 beats per minute



110 BPM

Ready

Start

Steps

- 1 Call for emergency help before starting.
- 2 Lay the person on their back on a firm surface.
- 3 Kneel beside their chest, place the heel of one hand on the center of the chest.
- 4 Place your other hand on top and interlock fingers.
- 5 Push hard and fast — at least 2 inches deep, 100–120 compressions/min.
- 6 Let the chest fully recoil between



Home



CPR Guide



Profile

Report an Emergency

Choose what's happening — we'll guide you instantly.



Injury / Fall



Bleeding



Burn



Choking



Cardiac / Chest
Pain



Other

Describe what happened (optional)

e.g. Person fell off a bike, minor
bleeding on the arm

Continue



Home



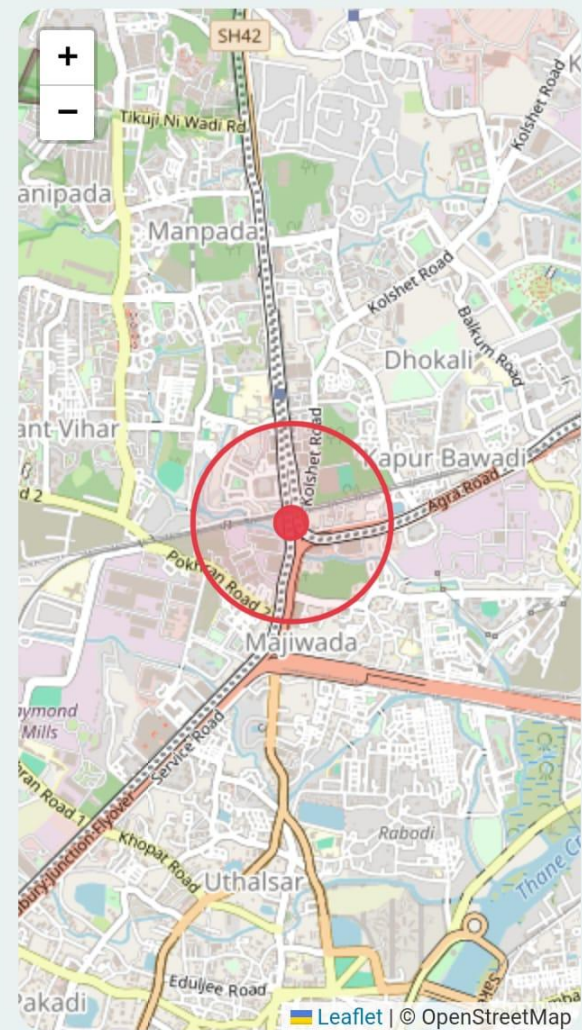
CPR Guide



Profile

Incident Location

500 m response radius



0 verified responder(s) within 5 km



Home



CPR Guide



Profile

Profile

himeshpal658@gmail.com

Role: Citizen

Language

English

English

Hindi

हिन्दी

Marathi

मराठी

Sign Out



Home



CPR Guide



Profile

Implementation Timeline

Weeks 1–4

Data Curation & Alignment

Annotate Roboflow datasets (wounds, burns, CPR); chunk WHO manuals into 300-token vectors via M3-Embedding

Weeks 5–8

Edge Quantization & Backend

Train YOLOv5s for edge inference; apply QLoRA + 4-bit GPTQ to the reasoning SLM for 2GB RAM compliance

Weeks 9–12

Geospatial & Multilingual Integration

Integrate translation matrices for rural dialects; build async multithreaded EMS alert routing

Weeks 13–16

Benchmarking & Field Testing

Monitor heap memory under thermal throttling; test TTFT speeds against disconnected real-world constraints

CONCLUSION

Closing the Golden-Hour Gap

SANKET bridges the research gap in mobile-first emergency informatics by directly countering the temporal compression that exhausts the golden hour. By moving away from vulnerable cloud LLMs to 4-bit quantized RAG pipelines running natively on 2GB edge devices, the system delivers zero-hallucination, localized triage guidance instantly — regardless of connectivity or infrastructure collapse.

Future Scope

- Unified on-device visual LLM (e.g. Phi-3.5-Vision) — single-pass reasoning
- Wearable IoT vitals ingestion (SpO2, heart rate) for dynamic prioritization
- National ambulance and hospital network integration for parallel mobilization

2GB

RAM ceiling

<5s

Target TTFT

94.5%

First-aid accuracy*

**from Pocket RAG (2026) reference benchmark*